Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Scuba Adventurers
Project / Solution Name	ExpiryGuardAI – Food Waste Prevention AI
Name (Leader)	Yap Sze Min
Email & Phone (Leader)	szemin3128@1utar.my, 0135293128
University Affiliation(s)	Universiti Tunku Abdul Rahman
Live Prototype URL	https://foodwastepreventionassistant.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Erinmin62/foodwastepreventionassistant.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

ExpiryGuardAI is a Replit-hosted web application. Owners use a React/Vite interface and sign in through Clerk. The interface sends requests to an Express/Node.js API, which verifies the owner and routes data to that owner's workspace. Pantry, menu, sales, planning, and automation records are stored in PostgreSQL as workspace-scoped data, accessed through Drizzle ORM. Ingredient photos are stored separately in private object storage and accessed through signed URLs.

Live planning and routine checks use application rules. For owner-reviewed recipe drafts and separate experimental forecast demos, the backend calls OpenAI through Replit's AI integration, validates the response, and returns it to the interface. Experimental forecast results are advisory and do not change live stock or place orders. When configured, a notification worker sends expiry emails through Resend.

System flow:

Owner → React/Vite interface + Clerk → Express API → PostgreSQL workspace data / private photo storage

AI features: Express API → Replit OpenAI integration → validated result → interface

Email alerts, when configured: notification worker → Resend → owner

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Main programming language for frontend and backend code	Apache-2.0
React 19.1.4	Build the web interface	MIT
Vite 7.3.6	Frontend development server and build tool	MIT
Express 5.3.0	Backend API server	MIT

Name & Version	Purpose in Project	License
Clerk React 6.17.8; Clerk Express 2.1.80	Sign-in and authentication	MIT
TanStack React Query 5.102.8	Fetches and caches frontend API data	MIT
Drizzle ORM 0.45.2	Connects backend code to PostgreSQL	Apache-2.0
node-postgres(pg)8.23.0	PostgreSQL database driver	MIT
Zod 3.25.76	Validates API data and AI responses	MIT
Tailwind CSS 4.3.3	Styles the web interface	MIT
Framer Motion 12.43.0	Animations	MIT
Recharts 2.15.4	Charts and data visualizations	MIT
Lucide React 0.545.0	Interface icons	ISC
Wouter 3.13.0	Frontend routing	Unlicense

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI API via Replit AI Integrations	Generate recipe drafts and clearly labeled experimental forecast demo. The forecast demo is separate from the app's live planning workflow	Replit-managed, usage-based; confirm whether usage is covered by your account or hackathon credits
Clerk API	Signs users in and authenticates requests to private workspaces.	Account/plan-dependent; confirm your Clerk plan
Resend API via Replit connector	Sends email notifications when the email worker and its settings are configured.	Account/plan-dependent; confirm Resend or Replit usage coverage
Replit App Storage	Stores uploaded ingredient photos and provides access to them.	Replit-provided; account/plan-dependent
Google Fonts API	Loads fonts used in the website interface.	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments — VM	Hosts the published application: the React/Vite website and Node/Express API. The prototype is currently published publicly on Replit.	VM deployment; exact machine plan and region need to be confirmed in Publishing settings
Replit PostgreSQL	Stores business workspaces and application records. Development and production databases are separate.	Replit-managed; production database is associated with the selected deployment region

Service & Provider	Purpose in Project	Plan / Region
Replit App Storage (backed by Google Cloud Storage)	Stores uploaded ingredient photos and other app files. The app accesses it through Replit's managed storage service.	Replit-managed; storage is collocated with the selected deployment region
Clerk	Provides sign-in and authentication for owner accounts and private workspaces.	Clerk-hosted; account plan and service region not confirmed

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GPT-5 mini	OpenAI, accessed through Replit AI Integrations	Pre-trained hosted model. Used to draft recipes and for separate experimental forecasts: a synthetic-data demo and an optional forecast from archived sales/preparation records after owner review and consent. The app does not fine-tune or train this model.	Proprietary; OpenAI and Replit AI Integrations service terms apply. Not open-source.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic demo and test records: sample ingredients, menu/recipes, sales, preparation, leftovers and expiry details	Generated within the project; no external source	Synthetic	Project-generated; no external dataset license	N
Owner workspace records: business profile, pantry and expiry details, menu/recipes, sales, preparation, leftovers, decisions, alerts and activity history	Entered or imported by each business owner into their private workspace	Self-collected / private	Owner-provided business data; used within the app with the owner's permission, not for redistribution	**Y, potentially** — records are linked to an owner account; business records are not intended to contain sensitive personal data

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Consented owner-history AI forecast input: archived sales and preparation records, including dates, item names, portions and leftovers	Selected from an owner's archive after the owner reviews the preview and gives one request consent; sent to OpenAI for an experimental forecast	Self-collected / private	Owner-provided data; use requires owner permission	**Y, potentially** — the request omits direct account identifiers and email, but records may remain identifiable through business context
Local held-out forecast evaluation records: historical sales and leftover portions	Evaluated locally from workspace records when an owner runs the check; not sent to the AI model	Self-collected / private, or synthetic in demo / test use	Owner-provided data with permission, or project-generated synthetic test data	**Y, potentially** for owner records; **N** for synthetic test records
Clerk account data: sign-in identifier and account emai	Supplied by the user and managed by Clerk	Private / service-managed	Clerk service terms	**Y**

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT (OpenAI; exact model/version not recorded)	Writing or refining prompts	Minor, if it was only used for prompt wordin
Claude (Anthropic; exact model/version not recorded)	Writing or refining prompts	Minor, if it was only used for prompt wording
Replit Agent (Replit; underlying model varies)	Creating and revising code, design, presentation materials, scripts, and documentation; debugging and testing	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated outputs are checked against a fixed schema, and recipe ingredients must match the owner's pantry. Owners review, edit, or reject drafts and recommendations; AI cannot change stock, approve decisions, or place orders. Archived owner sales and preparation data is sent to the AI model only after preview and explicit consent. Automated tests use synthetic cases. Forecasts are labeled experimental and advisory; their accuracy and business impact have not been validated with a real-business pilot.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
SDC	Replit platform	Used to develop, preview, test, and host the ExpiryGuardAI prototype.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was essential for developing, testing, and hosting this prototype during the SDC project, but it is not essential to the food-waste workflow itself. The app's core features could run on another cloud platform. Replacing Replit would require setting up equivalent hosting, database, file storage, authentication, and service connections, so it is replaceable but would require migration work.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

ExpiryGuardAI stores Clerk account identifiers and owner-entered business records, including business profile, pantry, expiry dates, menus, sales, preparation, leftovers, decisions, alerts, and activity. Ingredient photos are stored in private object storage. Access requires authentication, and records are scoped to the verified owner's workspace. Synthetic demo forecasts use generated data. The optional owner-history AI forecast sends dates, item names, and sales/preparation quantities to OpenAI via Replit only after preview and explicit consent; it excludes account IDs and email. If email alerts are enabled, recipient addresses and alert content are sent to Resend. The prototype has no fixed automatic retention period or whole-account deletion schedule. Some records, archive batches, and photos have individual delete controls; other data may remain until administratively removed. Third-party retention follows provider terms, and no additional retention period has been verified or configured.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

GPT-5 mini may hallucinate ingredients or produce inaccurate forecasts; its outputs have not been validated against real business outcomes, and no formal bias evaluation has been completed. The AI features also depend on OpenAI through Replit, so provider outages, rate limits, or service-term changes may affect availability.

Mitigations include validating outputs against strict schemas, rejecting recipe ingredients not present in the owner's pantry, and limiting AI requests. Owner-history records are sent only after preview and explicit consent. AI results are labeled experimental and advisory; owners must review suggestions, and AI cannot change stock, approve decisions, or place orders. Invalid or failed responses are rejected without changing operational records. Live expiry, demand-planning, and reorder functions use rules rather than AI; their real-world effectiveness also remains unvalidated.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

23 October 2026

[Date]